Tab 1

Conference Presentation Planning

Structure,

Problem space (1 minute)

- Helping bring

Moksh Side-

May → Sub section of the larger problem

Email for Gill

Mid Year Technical Report + Conference Presentation Planning

Mid Year Technical Report Overview / Sections

Moksh	
May	1. **Introduction** (Main problem and subproblem) **2. Development Methodology** a. ***Project Shift*** (choosing to focus on the facilitator - student side of the accessibility problem and trying to increase the quality of the education and reduce the workload of the 'teacher') b. ***Requirements Analysis*** (final criteria to assess the system's success) c. I***terative Development Process*** (prototype development w/ supervisor feedback) d. ***System Design Decisions*** (changes with assumptions, choosing to focus on reading and literacy and separation with ability to learn) 3. **System Architecture** (system flow, student side only has questionnaire, facilitator gets all the analysis of the individual students and the overall classes too) 4. **Learner Activities** (intended workflow and implementation for students, how often / frequent) 5. **Behaviour Collection** (Interactions collected and how to assess them, using studies and information to dictate which categories to put students in) 6. **Behaviour Feature Engineering** (explanation on how the raw events become behavioural insights and categorised for the pipeline to process) 7. **Recommendation Pipeline** (explanation of the rule-based implementation and why the decision tree model was used VS other models, explaining the measurements and then assessing them together) 8. **Current Progress & Future Work** (quick overview of the application, explaining how other subjects and modules can be used in a similar way, with other behaviour tracking features too, like eye tracking, etc. How facilitators will use the information, giving them a methodology)

Tab 3

Slide 1 — Title (~20s)

Thanks Moksh, I'll take it from here — just to remind you, I'm May. My side of this research project focuses more on the facilitator-students side of the problem and asks how we can better support less-experienced facilitators in understanding how their learners are actually progressing, so they can make more informed decisions.

Slide 2 — Motivation (~35s)

So as Moksh explained, our goal is to support less-experienced facilitators in classrooms to make education more accessible. One challenge is that facilitators often see outcomes of assessments, but not the learning process. A score tells you whether a learner answered correctly, but not how they reached that answer.

Recognising meaningful learning patterns takes experience, and research suggests this is challenging for novice educators and therefore the facilitators. Without that insight, facilitators may interpret the same learner very differently. That gap is what I wanted to address — which brings me to my research question.

Slide 3 — Research Question (~20s)

Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?

This is a proof-of-concept study — my focus is on demonstrating that the proposed framework can be implemented technically rather than evaluating it in a classroom environment.

Slide 4 — Framework Pipeline (~50s)

To investigate this question, I developed the framework shown here.

It begins with a structured learning activity that provides a consistent way of observing every learner. During the activity, the system captures not only the learner's answers, but also interaction data such as response timing, perceived confidence, hint usage and answer revisions.

This data is analysed to identify learner patterns and generate evidence-linked recommendations that are directly traceable to the observed interactions.

Facilitators can then use these insights to make more informed decisions. Importantly, the system supports facilitator judgement rather than replacing it.

Slide 5 — Prototype Scope & Design Assumptions (~55s)

The proof of concept here is built with Next.js and TypeScript, using strongly-typed data models for things like student responses and generated recommendations — which keeps the pipeline consistent from data capture through to the dashboard. The recommendation logic itself is a rule-based TypeScript engine, running directly against that data. Right now storage is browser-based for speed of iteration; moving that to a proper backend is the next infrastructure step.

Because this is a proof of concept, I deliberately limited the scope to a single literacy activity.

I chose a target leaner age group of Year 3–6 represents the transition from learning to read to reading to learn, which helps facilitators understand whether a leaner needs help understanding the task or how to read. This is why the proof of concept was designed to focus on comprehension because it's key to any kind of teaching.

Recommendations are intentionally advisory, they highlight patterns but never diagnose learners or replace facilitator judgement.

Activities are designed to be completed regularly in around 5–10 minutes, allowing learner progress to be tracked over time. And so there’s a baseline to refer to.

Students complete a short reading comprehension task before class. During the activity, the system captures interaction data such as response timing, confidence, hint usage, answer revisions and assessment performance. These interactions become the input to the analytics pipeline.

On the facilitator side, this information is transformed into a learner summary that combines performance with interaction behaviours. The system generates evidence-linked recommendations while allowing facilitators to monitor both individual progress and class-wide trends.

Currently Each feature is evaluated against a set of predefined rules, and any rule that's triggered generates a behavioural indicator paired with the exact interaction data that caused it to fire. If multiple rules trigger, the facilitator sees all of them, not a single collapsed judgement. That's what makes the output fully traceable — every recommendation on the dashboard can be followed back to something the learner actually did.

Slide 6 — Contributions & Future Work (~45s)

This research contributes in three ways.

Educationally, it investigates how facilitators can be better supported and how the proposed framework can reduce the load on them of keeping track of every student to make better decisions.

Technically, it proposes a general decision-support framework that is not limited to literacy.

From a research perspective, it explores how learner interaction data can be used to inform facilitator decision making.

My next step is to compare the current rule-based recommendation engine with machine-learning approaches. Beyond the project, the modular pipeline means the same framework can be applied to different learning activities, subjects and age groups, allowing for this project to be utilised by almost any area and used to help people with different teaching experience levels.

Q: “How is this different from existing products?”

Key points

- Not another learning platform.
- Complements existing systems (e.g., Google Classroom, Moodle).
- Existing platforms focus on **content delivery** and **assessment outcomes**.
- This focuses on **facilitator decision support**.
- Analyses **how** learners complete activities, not just the final score.
- Uses interaction data:
    - Response time
    - Confidence
    - Hint usage
    - Answer revisions
- Produces **evidence-linked recommendations**, not just analytics.
- Contribution = **general decision-support framework**.

Q: “Why didn't you just use AI?”

Key points

- AI is **future work**, not a research contribution.
- The goal is to validate whether interaction data contains meaningful signals.
- Rule-based system provides:
    - Transparency
    - Explainability
    - Traceability
- Every recommendation links directly to learner interactions.
- Establish baseline first → compare AI later.

Q: “If AI is your future work, why not start with AI?”

Key points

- Need to validate the data first.
- Unknown whether interaction data is meaningful.
- Rule-based baseline makes evaluation easier.
- No representative labelled dataset yet.
- AI comparison comes after the baseline is established.

Q: “How do you know the recommendations are correct?”

Key points

- I don't claim they're educationally correct.
- Proof-of-concept demonstrating technical feasibility.
- Shows interaction data **can** become recommendations.
- Educational validation requires classroom studies.
- Future work with facilitators and learners.

Q: “How would you evaluate success?”

Current evaluation

- Technical feasibility.
- Data captured correctly.
- Data analysed correctly.
- Evidence-linked recommendations generated.

Future evaluation

- User studies.
- Identification accuracy.
- Consistency.
- Decision-making efficiency.
- Facilitator confidence.

Q: “How does your rule-based implementation work?”

Pipeline

- Collect learner interaction data.
    - Accuracy
    - Confidence
    - Response timing
    - Hint usage
    - Answer revisions
    - Passage revisits
- Extract behavioural features.
- Evaluate predefined behavioural rules.
- Trigger behavioural indicators.
- Generate evidence-linked recommendations.
- Display to facilitator.

Rule design

- Each rule represents a behavioural pattern.
- Rules can trigger independently.
- Multiple indicators may appear simultaneously.
- Every recommendation references the evidence that triggered it.
- Transparent and explainable.

Current limitations

- Small rule set.
- Proof-of-concept only.
- Rules demonstrate framework feasibility.
- Can later be refined or replaced with ML.

One-sentence summary

> The system extracts learner interaction features, evaluates predefined behavioural rules, and generates evidence-linked recommendations where every recommendation is directly traceable to the observed learner interactions.

Research contribution (if asked)

Key points

- Not the literacy activity.
- Not the dashboard.
- Not AI.

Contribution is:

- General learner analytics framework.
- Decision-support for facilitators.
- Evidence-linked recommendations.
- Literacy is the proof-of-concept domain.
- Framework intended to be transferable to other subjects.

Questions:

- "How is this different from existing products?"

Most existing educational platforms focus on delivering content and reporting assessment outcomes such as scores or completion rates. My research focuses instead on supporting facilitator decision making by analysing learner interactions during an activity. Rather than only reporting whether a learner answered correctly, the system captures how they completed the task for example through response timing, confidence, hint usage and answer revisions and transforms those observations into evidence-linked recommendations. The contribution is not another learning platform, but a decision-support framework that can sit alongside existing learning activities

Rather than replacing existing learning platforms, this framework is intended to complement them by analysing learner interactions and providing facilitator-focused decision support

Some educational platforms are beginning to integrate AI, but much of that work focuses on content generation, tutoring or automated feedback. My research is specifically investigating decision support for facilitators using learner interaction data. The current prototype establishes an explainable baseline that future AI approaches can be compared against.

- “Why didn’t you just use AI?”

The aim of this research is to investigate whether learner interaction data can be transformed into meaningful facilitator recommendations. A rule-based system provides complete transparency because every recommendation can be traced back to the observed learner interactions. That makes it easier to evaluate the framework before introducing more complex AI models. It is intended to use AI but I wanted to lay the groundwork first.

AI isn't the research contribution, learner analytics for facilitator decision support is. The current rule-based system provides an explainable and traceable baseline that lets me evaluate whether learner interaction data contains meaningful signals. Once that baseline is established and sufficient data has been collected, AI models become a natural next step for comparison.

- “If AI is your future work, why not start with AI?”

Because before evaluating different AI approaches, I first need to establish that the underlying data being collected is meaningful. This research asks whether learner interaction data contains useful signals for facilitator decision support. The rule-based implementation provides a baseline that allows those signals to be validated before introducing more complex models.

- “How do you know the recommendations are correct?”

At this stage, I don't claim that the recommendations are educationally correct. This research is a proof of concept demonstrating that learner interaction data can be transformed into evidence-linked recommendations. Validating the educational effectiveness of those recommendations is future work involving classroom studies.

- “How would you evaluate success?”

The current project evaluates technical feasibility: whether learner interaction data can be collected, analysed, and converted into evidence-linked recommendations. Future evaluation would involve user studies measuring whether facilitators identify learners requiring support more accurately, consistently, or efficiently when using the system.

- “How does your rule-based implementation work right now?”

The current implementation follows a deterministic rule-based pipeline. First, the system collects learner interaction data during the activity, including measures such as accuracy, confidence, response timing, hint usage, answer revisions, and passage revisits.

These interaction features are then evaluated against a set of predefined rules. Each rule represents a specific behavioural pattern that may warrant facilitator attention. For example, frequent hint usage combined with multiple passage revisits may indicate that a learner is struggling to interpret the task, even if they eventually answer correctly.

If a rule is triggered, the system generates a corresponding behavioural indicator and an evidence-linked recommendation. Importantly, the recommendation isn't generated in isolation—it always includes the learner interactions that caused that rule to fire.

If multiple rules are triggered, the system can surface multiple indicators, allowing the facilitator to see all relevant patterns rather than a single overall judgement. The output is therefore transparent and explainable, with every recommendation traceable back to observable learner behaviour.

At the moment, the rule set is intentionally small because this is a proof-of-concept. Each rule was designed to demonstrate how different learner interaction patterns can be translated into evidence-linked recommendations. The intention isn't to claim these are exhaustive, but to establish a baseline that can later be refined or compared with machine-learning approaches.

The system extracts learner interaction features, evaluates them against predefined behavioural rules, and generates evidence-linked recommendations where every recommendation can be traced directly back to the observed learner interactions.